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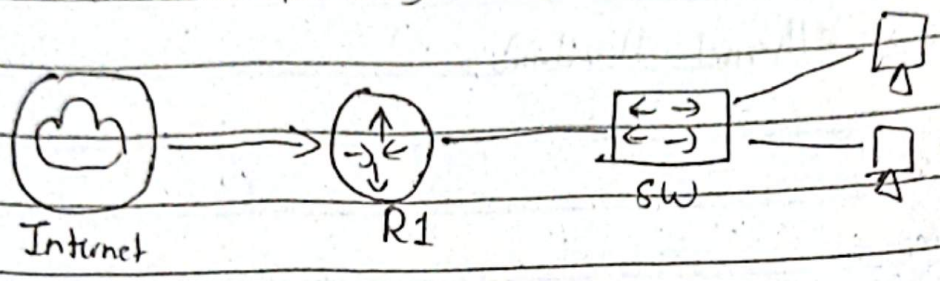
Ethernet LAN Switching (Part 1)
(How data transfer)



OSI Model - Physical Layer

① Physical Layer

→ Physical Layer defines physical characteristics of the medium used to transfer data between devices.
For example: voltage levels, maximum transmission distances, physical connectors, cable specifications etc.

② Data Link Layer

↳ Data Link Layer provides nodes-to-nodes connectivity and data transfer. (For example, PC to switch, switch to router, router to router).

↳ This defines how data is formatted for transmission over a physical medium. For example Copper UTP cables).

↳ Detects and (possibly) corrects Physical Layer error.
↳ Uses Layer 2 addressing, seperate from Layer 3 addressing
→ Switches operate at Layer 2.

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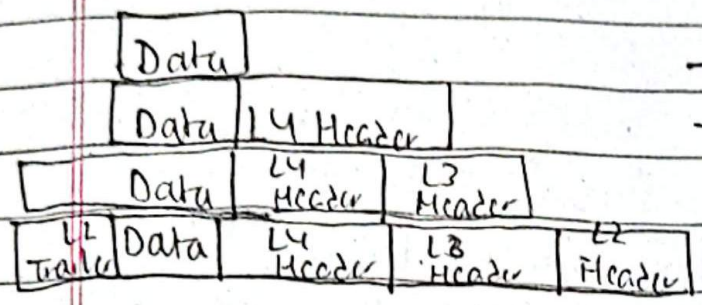
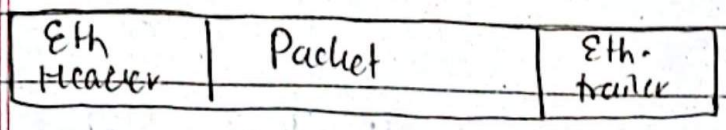
PDU's - Encapsulation process

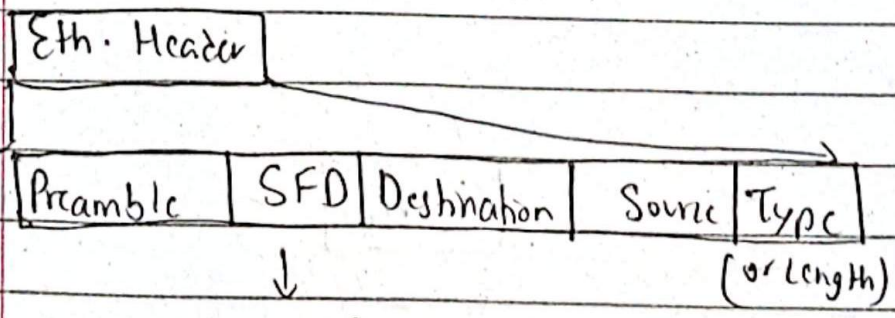


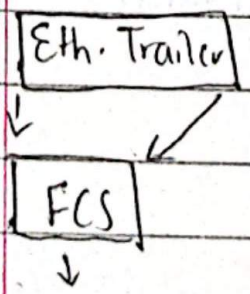
- Data
- Segment
- Packet
- Frame

} Protocol Data Units (PDUs)

Ethernet Frame







Frame Check Sequence

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Preamble

SFD

- Preamble
→ Length : 7 byte (56 bits)
→ Allows device to synchronize their receiver clock.
→ 10101010 *7

- SFD
→ 'Start Frame Delimiter'
→ Length : 1 byte (8 bits)
→ Marks the end of the preamble and the begining of the rest of the frame.
→ 10101011

Destination

Source

→ ~~Des~~ Indicates the device sending & receiving the frame
→ Consists of the destination and source 'MAC' address
→ MAC address = 6 byte (48 bits)

Type or Length

→ 2 bytes (16 bits)
→ Used to represent type of packet or LENGTH based on the LENGTH of packet.

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- A value of 1500 or less in this field indicates the LENGTH of the ~~encapsulation~~ encapsulated packet (in bytes)
- A value of 1536 or greater in this field indicates the type of the encapsulated packet usually ~~a~~ 1 (IPv4 or IPv6) and the length is determined through other methods.

IPv4 = 0x0800 (Hexadecimal) = 2048 in decimal
IPv6 = 0x086DD (Hexadecimal) = 34525 in decimal

Eth. Trailer

FCS
- Frame Check Sequence
- 4 bytes (32 bits) in length.
- Detects corrupted data by running 'CRC' algorithm over the received data

MAC Address

- 6-byte (48 bit) physical address assigned to the device when it is made
- AKA BIA ('Burned-In Address')
- Is globally unique
- The first 3 bytes ~~of~~ are the OUI (Organizationally Unique Identifier) which is assigned to the company making the device
- The last 3 bytes are the unique to the device itself
- It is written as 12 hexadecimal characters

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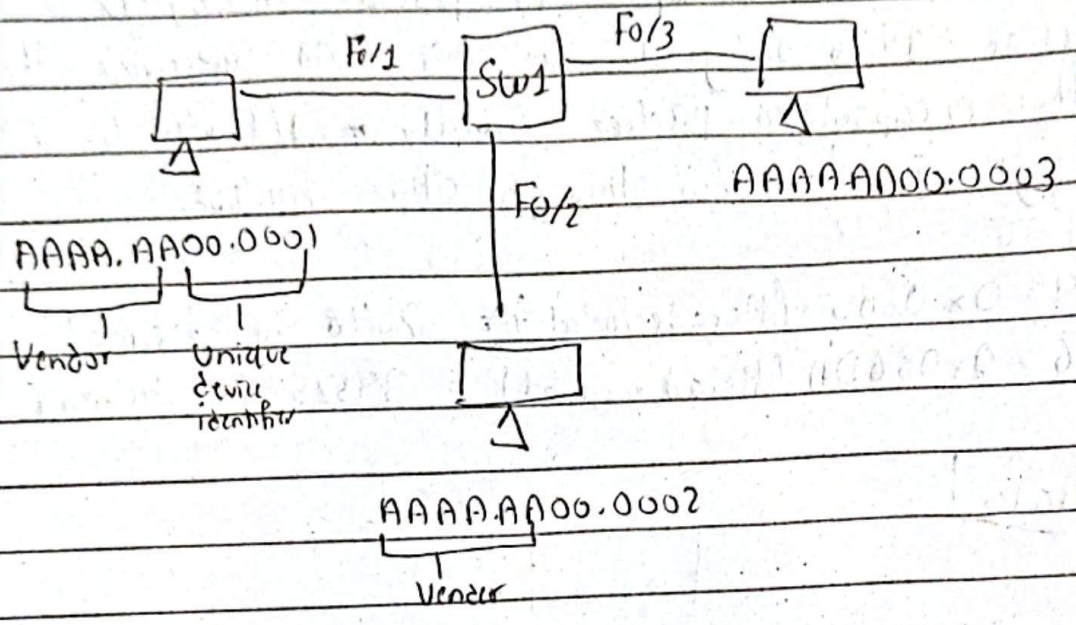
Decimal



Ethernet LAN Switching (Sending traffic within a LAN)

Preamble	SFD	Destination	Source	Type	FCS
7	1	6	6	2	4

→ Preamble and SFD is usually not considered part of Ethernet header.

→ Therefore the size of the Ethernet header + trailer is 18 bytes.

~~(6+6+2)+4~~ 6 + 6 + 2 + 4 = 18 bytes

→ Minimum size of ethernet = 64 bytes [Header + Payload + Trailer]

So, 64 bytes − 18 bytes ⇒ 46 bytes

∴ Minimum Payload (packet) is 46 bytes

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→ If the payload is less than 46 bytes, padding bytes are added i.e.

34 bytes ~~packet~~ packet + 12-byte padding = 46 bytes
[payload]

ARP (Address Resolution Protocol)

↳ ARP is a protocol that is used to discover the layer 2 address (MAC address) of a known layer 3 address (IP).

↳ It consists of 2 messages:-

① ARP Request ② ARP Reply.

- ARP Request is broadcast ⇒ Sent to all host on the network
- ARP Reply is unicast

arp -a ⇒ to view the ARP table

- IP address (layer 3 address)
- MAC address (Layer 2 address)
- Type static = default entry
- Type dynamic = learned through ARP

Ping

↳ A network utility that is used to test reachability

↳ Measures round-trip time

↳ Uses two messages: ICMP Echo Request
ICMP Echo Reply

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Commands

Sw# show arp

Sw# show mac address-table

Sw# clear mac address-table dynamic Imp

Sw# ~~show~~ clear mac address-table dynamic address 0000.0000.0012

↳ To remove specific address

Sw# clear mac address-table dynamic interface Gi 0/0

Imp